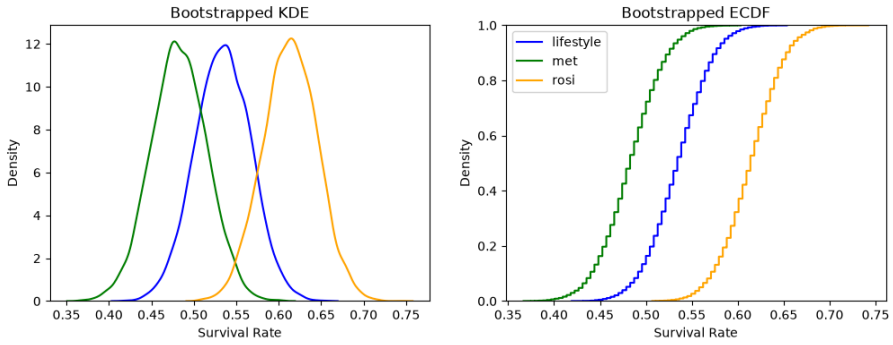
Lab 6

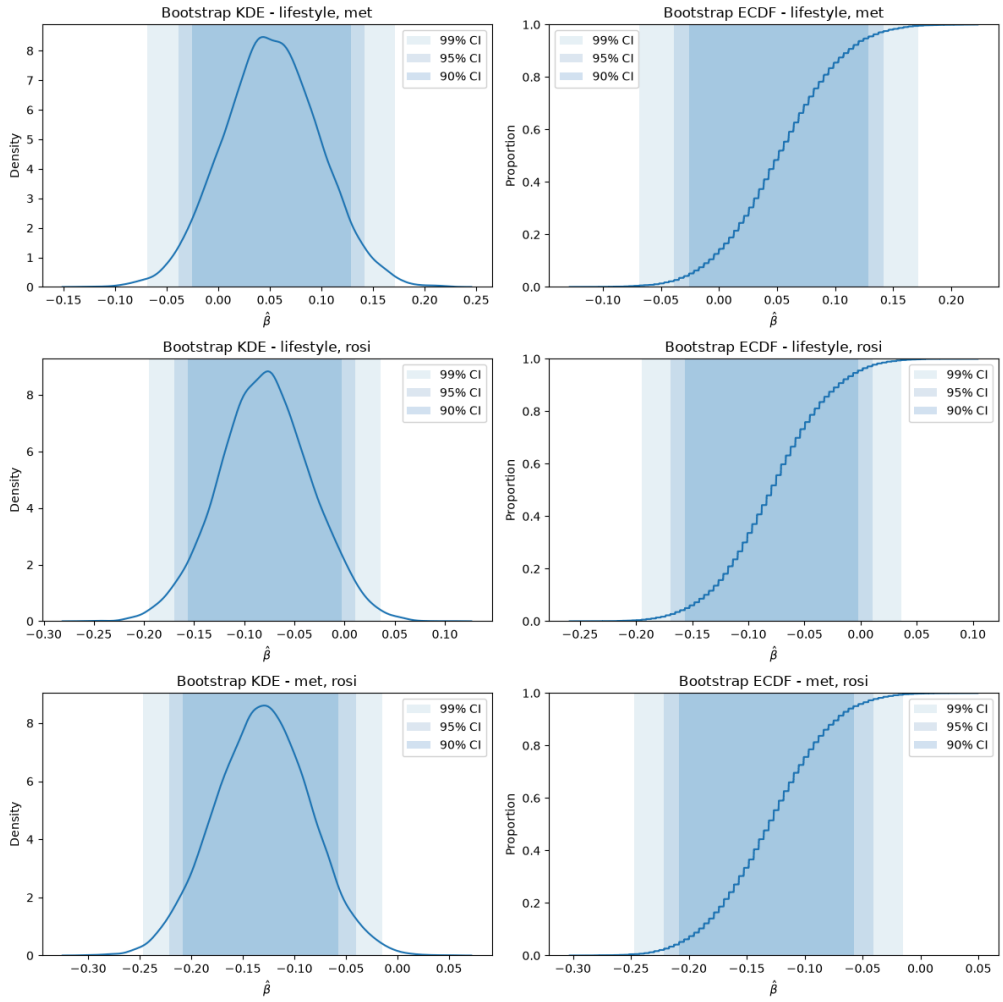
In this study, we looked at three treatments for pre-diabetes. The treatments used were metformin ("Met"), rosiglitazone ("Rosi"), and lifestyle adjustments ("Lifestyle"). For the experiment, 699 patients across the groups were tested to see if their treatment was a success. Based on these results, we can assess whether there is a significant difference in the effectiveness of the treatments.

Treatment	Failure	Success	Average
Lifestyle	109	125	0.534
Met	120	112	0.483
Rosi	90	143	0.614

The samples show that the Rosi treatment was the most successful in the trial, with a success rate of 61.4%. Following behind, the Lifestyle treatment had a success rate of 53.4%, and the Met treatment was the least successful at 48.3%. However, with the relatively small sample size, we cannot yet say with certainty that Rosi is the most effective. To measure this, we want to see how variable our success estimate becomes with different sample data. We can bootstrap our current sample, which allows us to estimate the distribution of the success rate across different samples.



The figure above shows the sampling distributions of survival rate for all three treatments. The left plot shows that each group has roughly the same variance in their sampling distribution, and each of these are normally distributed. The Rosi group has the highest mean, which matches with our results from the table. The ECDF plot on the right displays stochastic dominance for Rosi over the other groups, with its survival rate always being higher for each density quantile.



Subtracting our distributions also allows us to see the sample effect sizes for each pair of treatments. Confidence intervals for these effect sizes are shown in the plots above. Using an alpha of 0.05, we want to see if 0 is within the 95% confidence intervals. If 0 is not in the

interval, we can reject the null hypothesis that the groups have no difference. In this case, the only treatment pair where we can reject the null hypothesis is the Rosi-Met pair. Therefore, we can say that Rosi is significantly better at treating pre-diabetes than Met. However, we cannot say anything about how the lifestyle adjustments compare to either Rosi or Met. If we used an alpha of 0.1, we would actually find a significant difference between Rosi and Lifestyle, but we are using the standard of alpha = 0.05 to reduce the chance of a Type I error. Overall, we see that rosiglitazone was the most effective in the clinical trials, and we can safely say that it is significantly more effective than metformin.